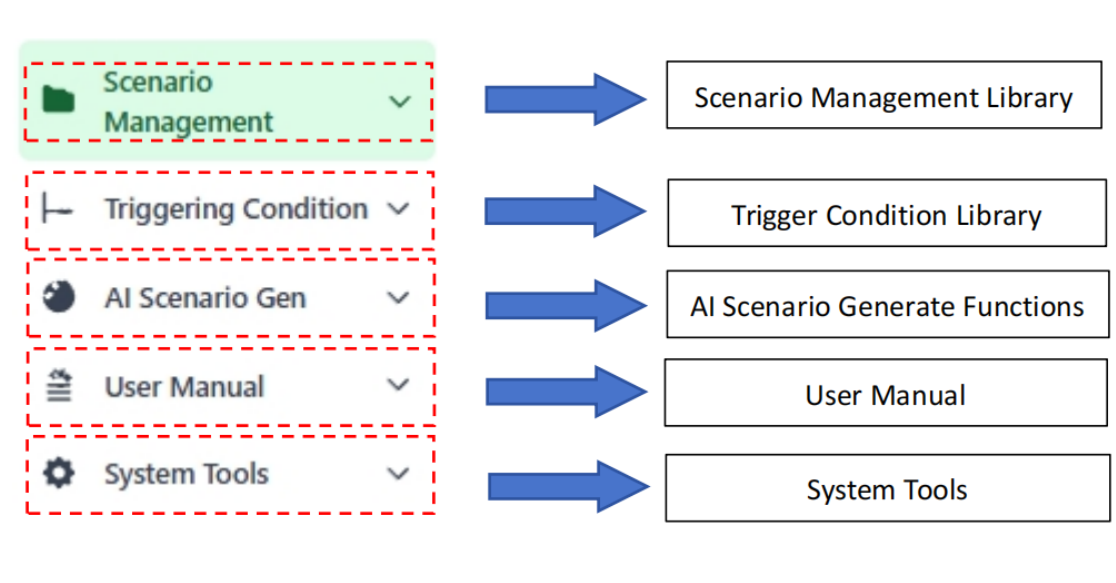
AI Data Engine Software User Manual

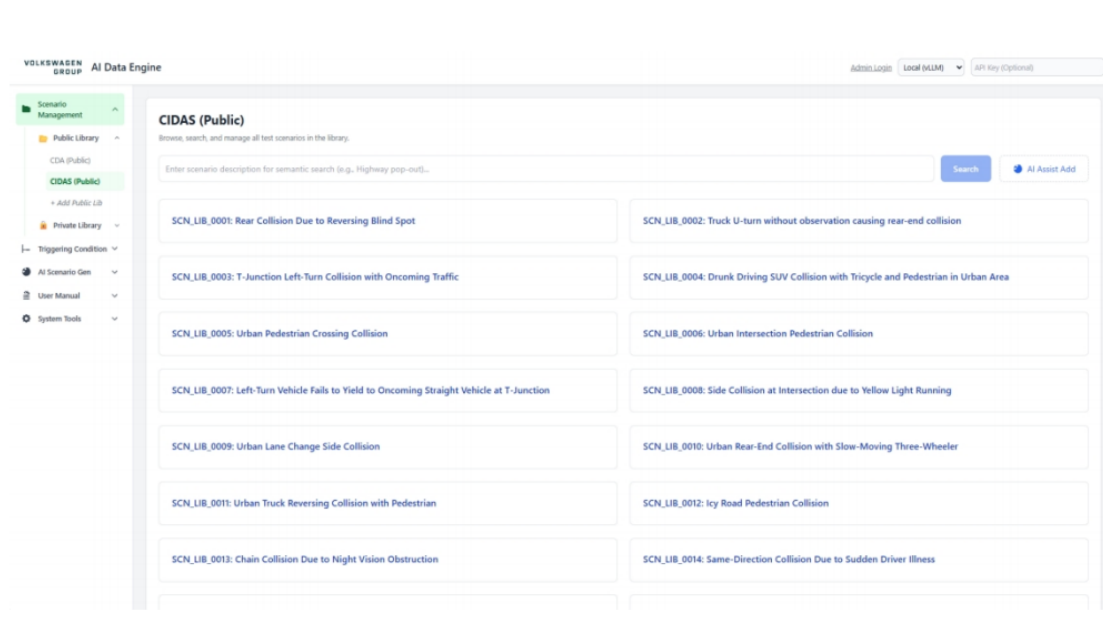
The software primarily consists of two major databases: a Scenario Management Library and a Trigger Condition Library. along with two key features: AI-assisted scenario generation and AI-driven scene creation based on user requirements.

- **Initial user interface display：**

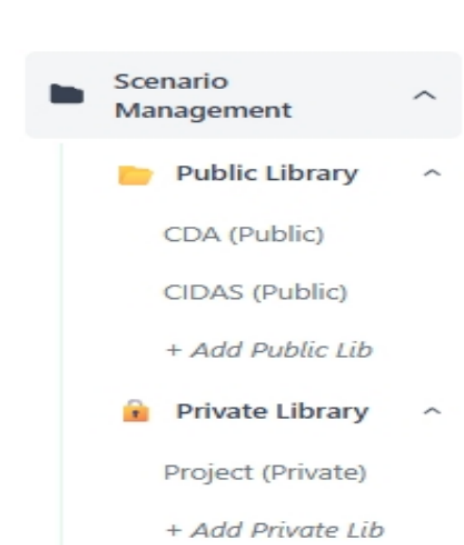

Scenario Management
Triggering Condition
AI Scenario Gen
User Manual
System Tools
Scenario Management Library
Trigger Condition Library
AI Scenario Generate Functions
User Manual
System Tools

1. Scenario Management Library Description:

Click on the Scenario Management Library to access the following interface:

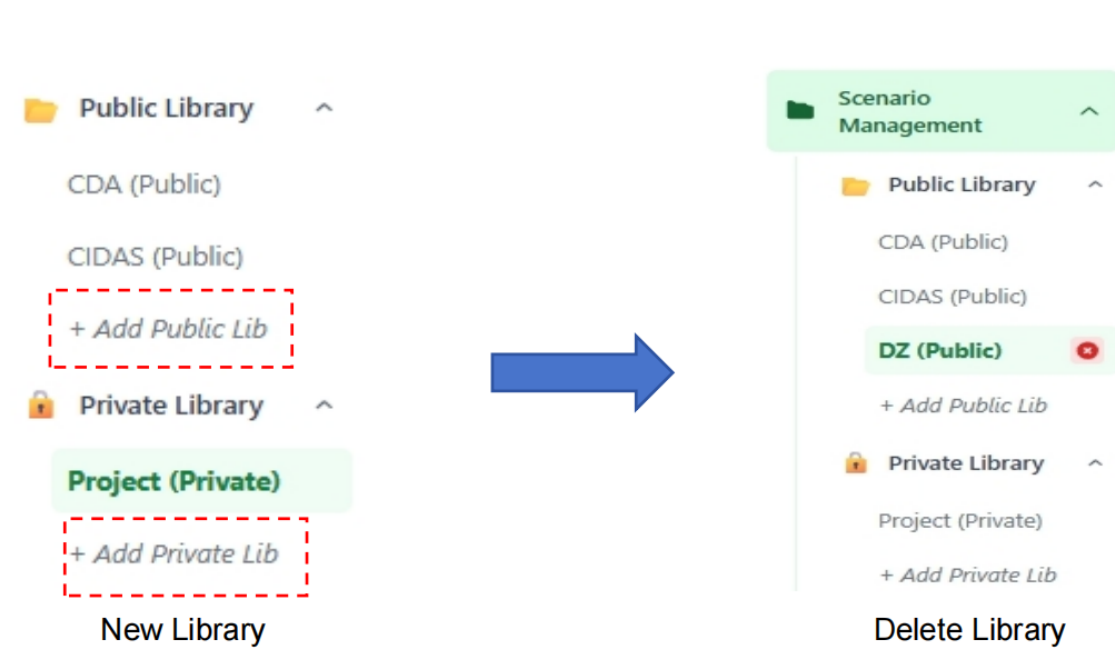

AI Data Engine
CIDAS (Public)

The Scenario Management Library primarily consists of a Public Library and a Private Library. The Public Library includes CIDAS scenarios and CDA scenarios. Within the Private Library, the Project Library contains generalized scenarios derived from accident data collected in Chongqing over the past five years by the ChongQing university Laboratory.


Scenario Management
Public Library
CDA (Public)
CIDAS (Public)
+ Add Public Lib
Private Library
Project (Private)
+ Add Private Lib

Process for Creating a New Public/Private Library:

Click the "Add" button in either the Public or Private Library to create a new public or private library. You can name it as needed, and for private libraries, an access password can be set. For any newly created library, you can click the delete button on the right to remove it.

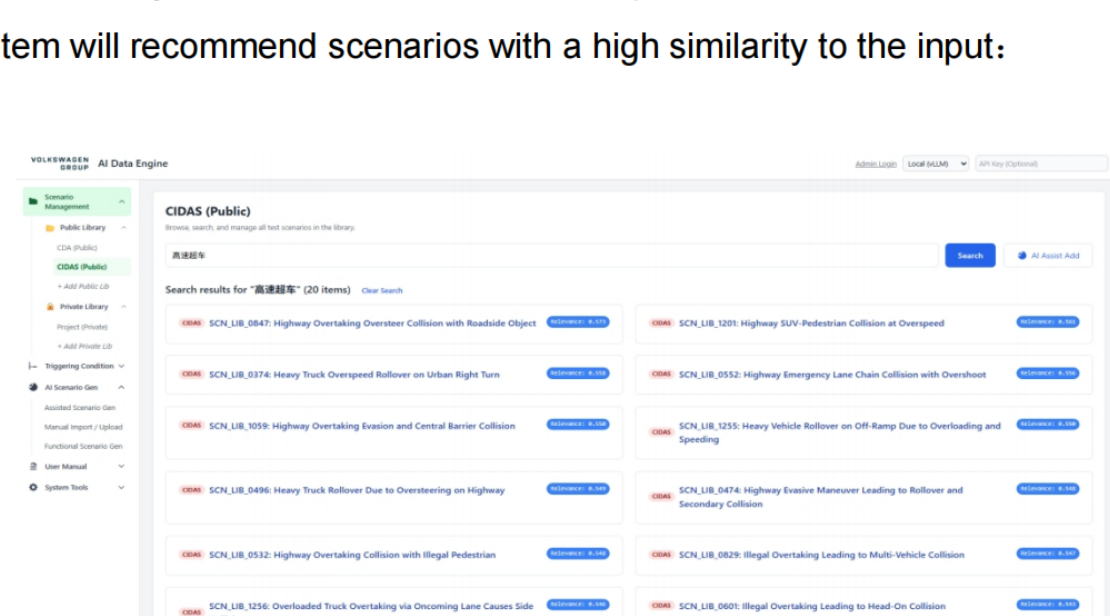

Public Library
CDA (Public)
CIDAS (Public)
+ Add Public Lib
Private Library
Project (Private)
+ Add Private Lib
Scenario Management
Public Library
CDA (Public)
CIDAS (Public)
DZ (Public)
+ Add Public Lib
Private Library
Project (Private)
+ Add Private Lib

New Library

Delete Library

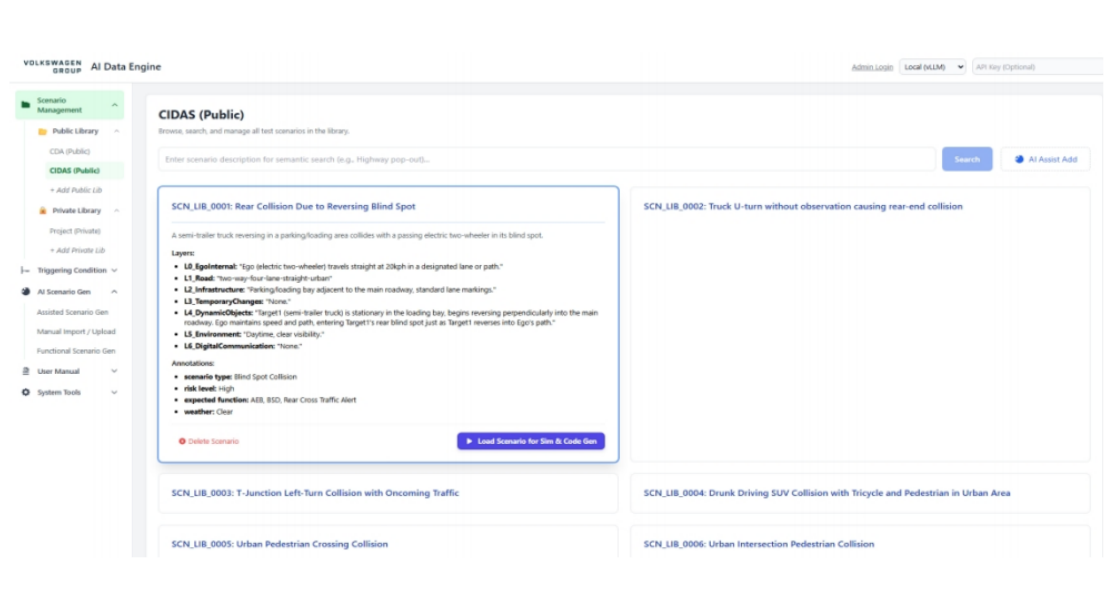
Similarity Search Process:

Enter any test function or related description, then click Search , The system will recommend scenarios with a high similarity to the input：

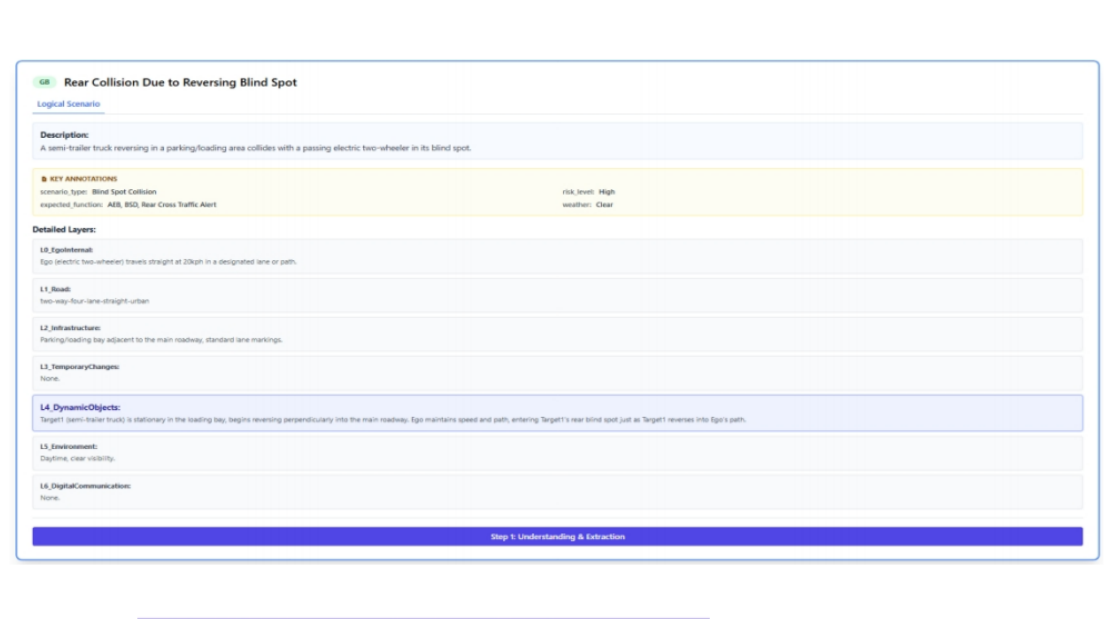

AI Data Engine
CIDAS (Public)
Search

Complete Simulation Scenario Process for Observing the Database:

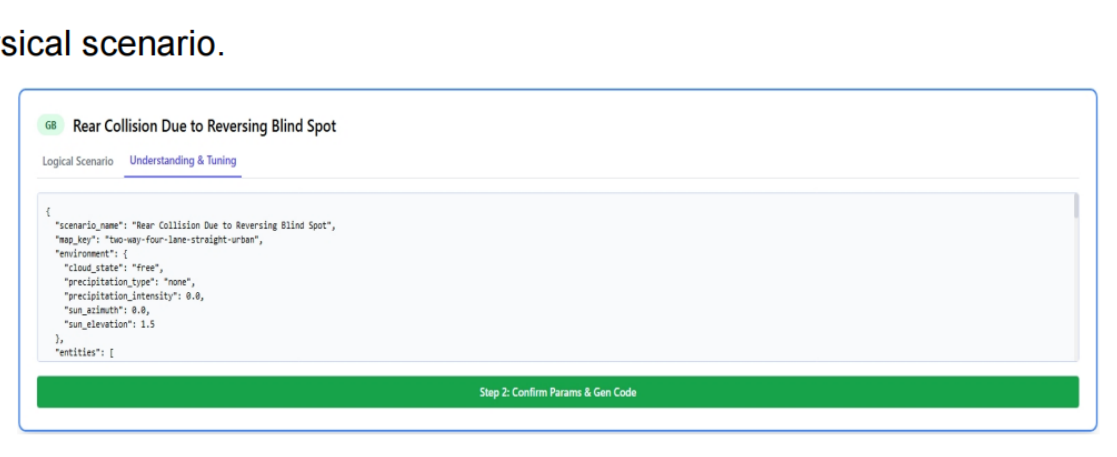
（1）Click on the corresponding database card to access the relevant data：


AI Data Engine
CIDAS (Public)

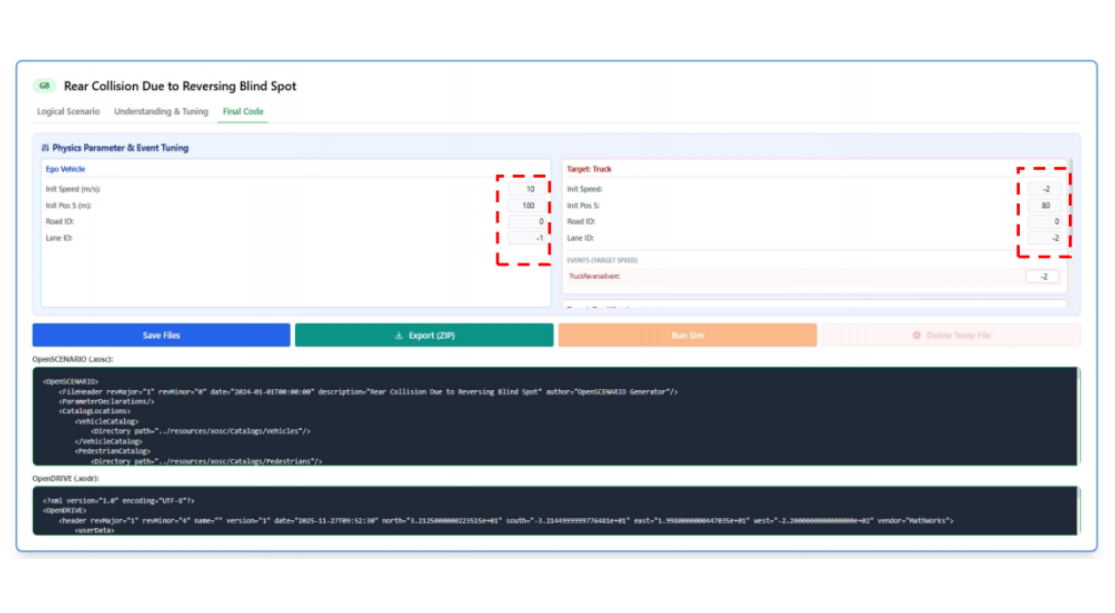
（2）Click "Load Scenario for Simulation & Code Generation" to access the following interface：


Rear Collision Due to Reversing Blind Spot
Step 1: Understanding & Extraction

（3）Click Step 1: Understanding & Extraction to access its corresponding physical scenario.

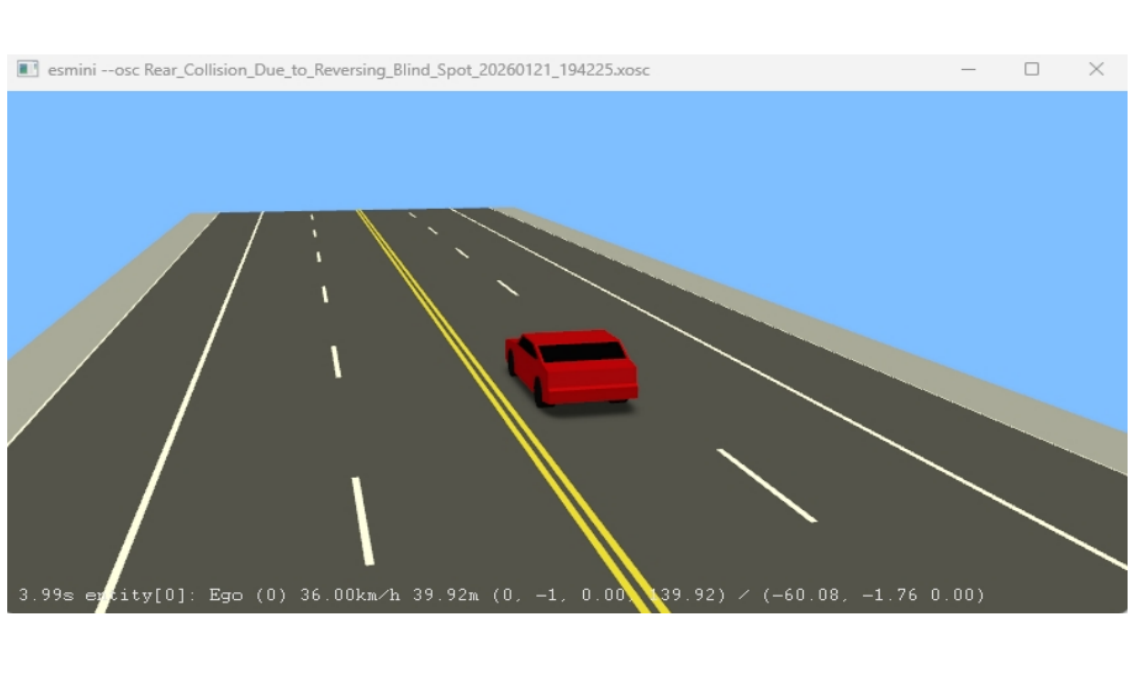

Rear Collision Due to Reversing Blind Spot
Logical Scenario
Understanding & Tuning
Step 2: Confirm Params & Gen Code

（4）Click Step 2: Confirm Params & Gen Code to view its corresponding OS code：

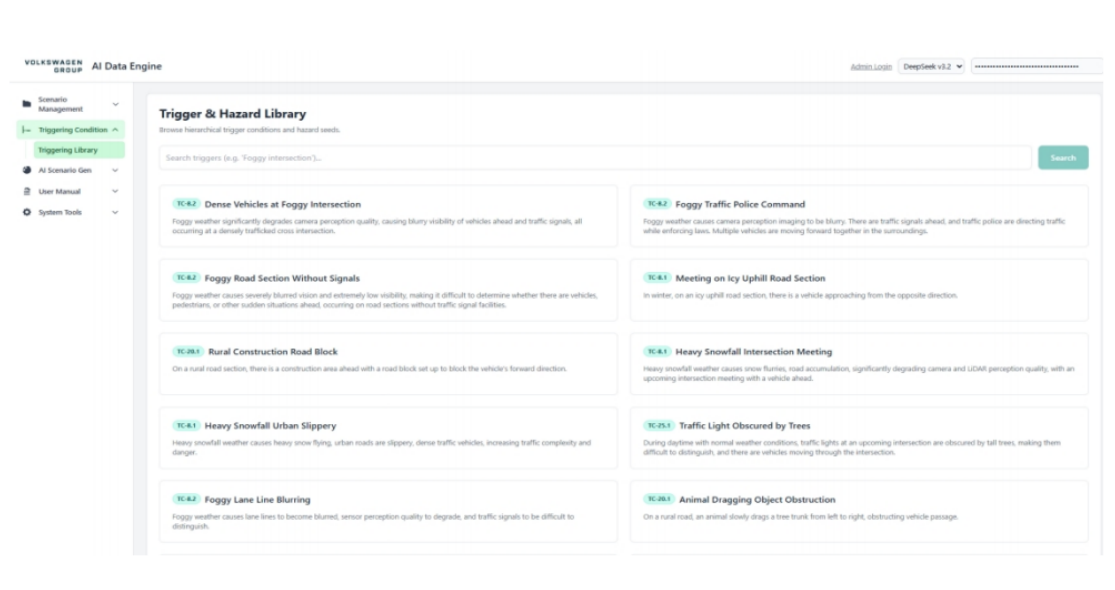

Rear Collision Due to Reversing Blind Spot
Logical Scenario
Understanding & Tuning
Final Code
Save Files
Export (ZIP)

Among these, the above data—including the speed and position of the ego vehicle and traffic participants—can be modified as needed。

（5）Click Save Files (Apply Params) to save the code, and click Esmini Simulation to start the simulation. To avoid data redundancy, after saving the code, you can click Delete Temp File to delete the saved code。


esmini --osc Rear_Collision_Due_to_Reversing_Blind_Spot_20260121_194225.xosc

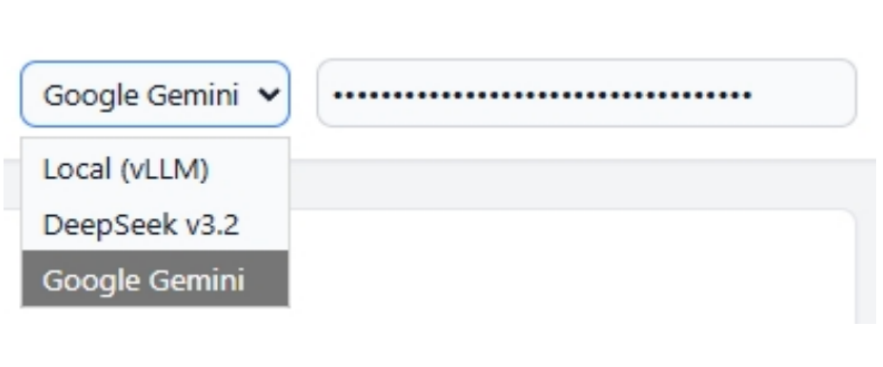
2.Triggering Condition Library：


AI Data Engine
Trigger & Hazard Library

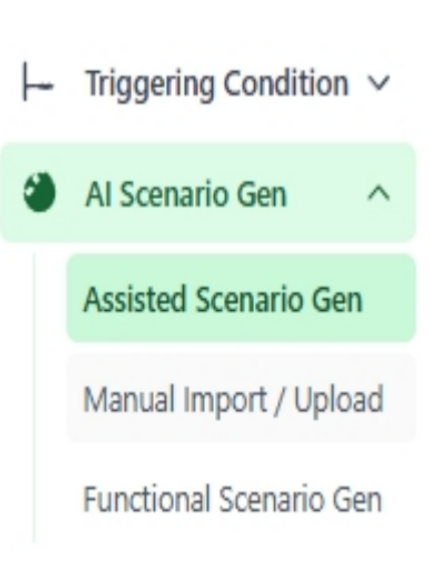
The Triggering Condition Library serves as a resource for AI to generate complex scenarios through layering and combination.

3. AI Scenario Generation Function：

When using AI features, you need to enter the corresponding model and its API information in the upper-right corner：


Google Gemini
Local (vLLM)
DeepSeek v3.2
Google Gemini

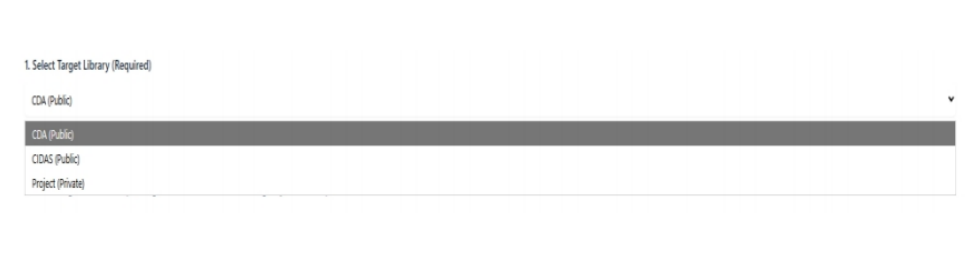
Its functions mainly consist of three parts: AI-assisted scenario import, AI scenario generation, and user-defined scenario addition：


Triggering Condition
AI Scenario Gen
Assisted Scenario Gen
Manual Import / Upload
Functional Scenario Gen

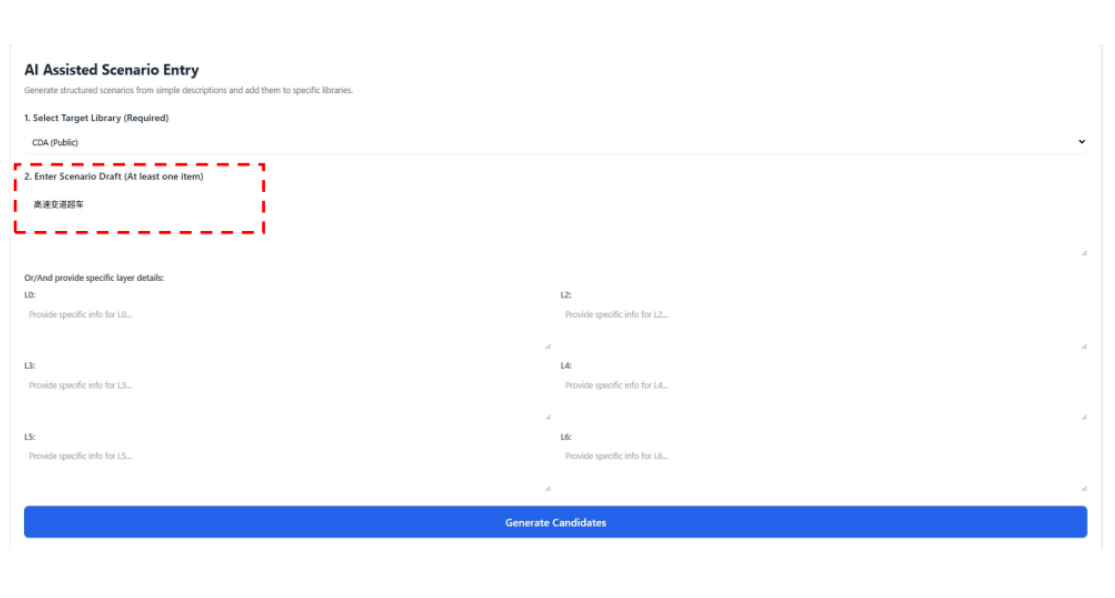
（1）AI-Assisted Scenario Import Process：

Click Assisted Scenario Gen to enter the AI-assisted scenario import function。

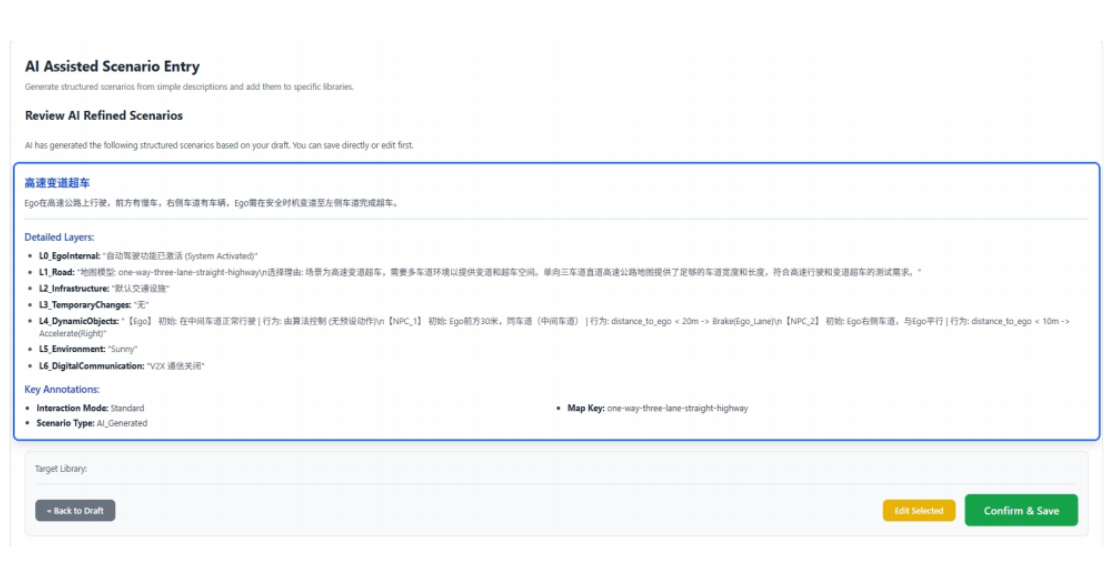
First, select the library you wish to add to：


1. Select Target Library (Required)
CDA (Public)
CIDAS (Public)
Project (Private)

In the scenario description field, you can provide a brief text description of the scenario. The L0 to L6 levels below are optional. If the user needs to specify certain details or data, they can fill in the relevant fields; otherwise, the AI will appropriately supplement and generate the scenario based on the description provided.


AI Assisted Scenario Entry
Generate Candidates

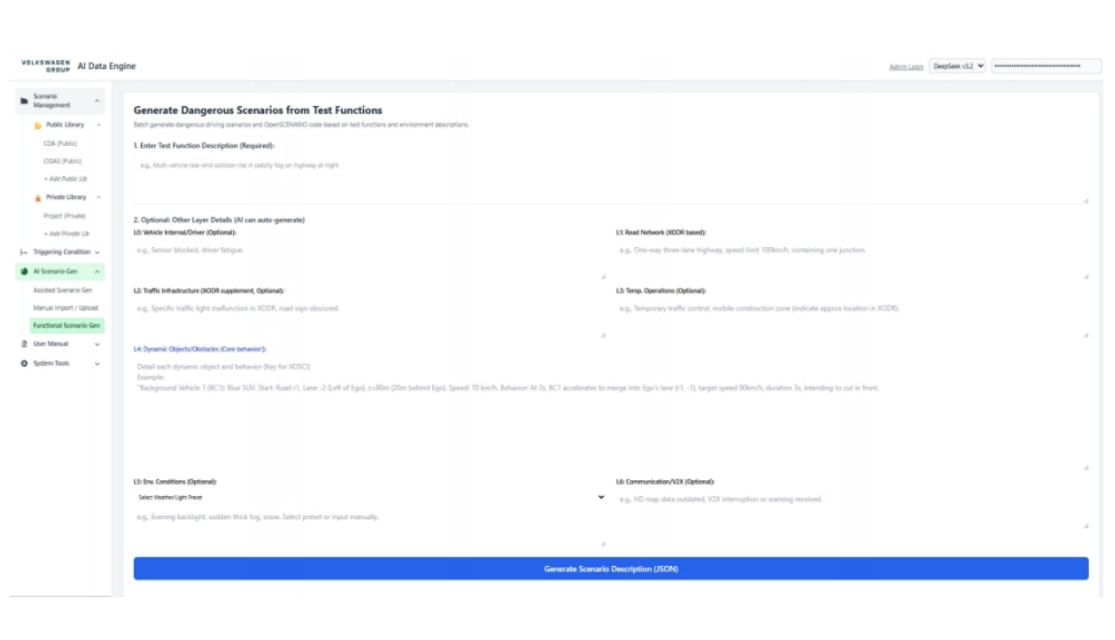
Then click Generate Candidates to complete the scenario。


AI Assisted Scenario Entry
Review AI Refined Scenarios
Edit Selected
Confirm & Save

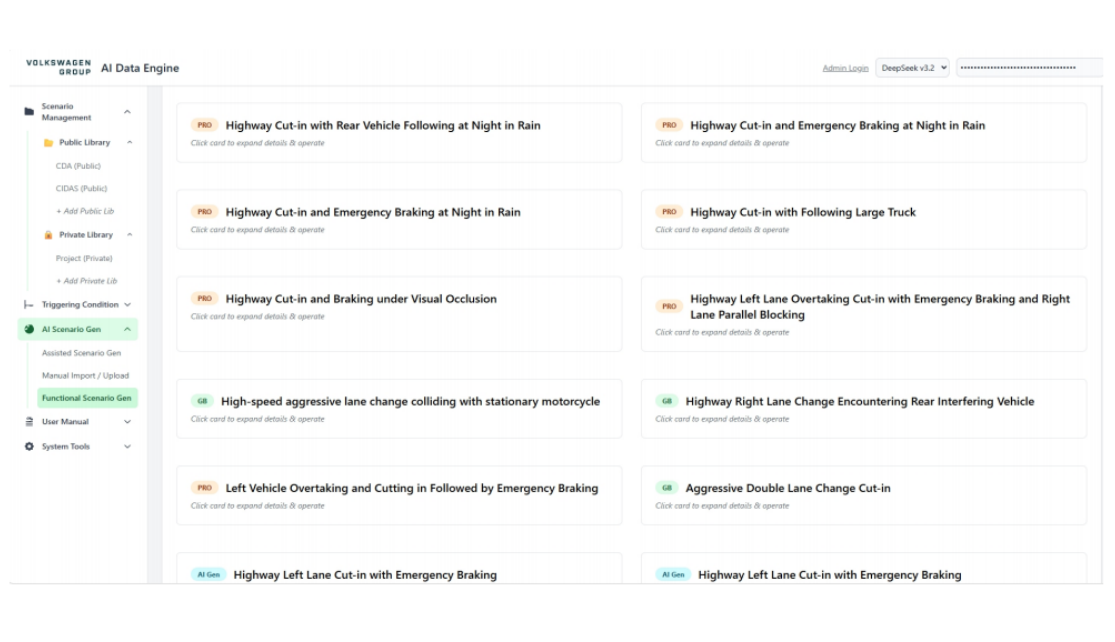
Finally, you can click Edit Selected to edit the document or click Confirm & Save to save the scenario。

（2）AI Scenario Generation Function Process：

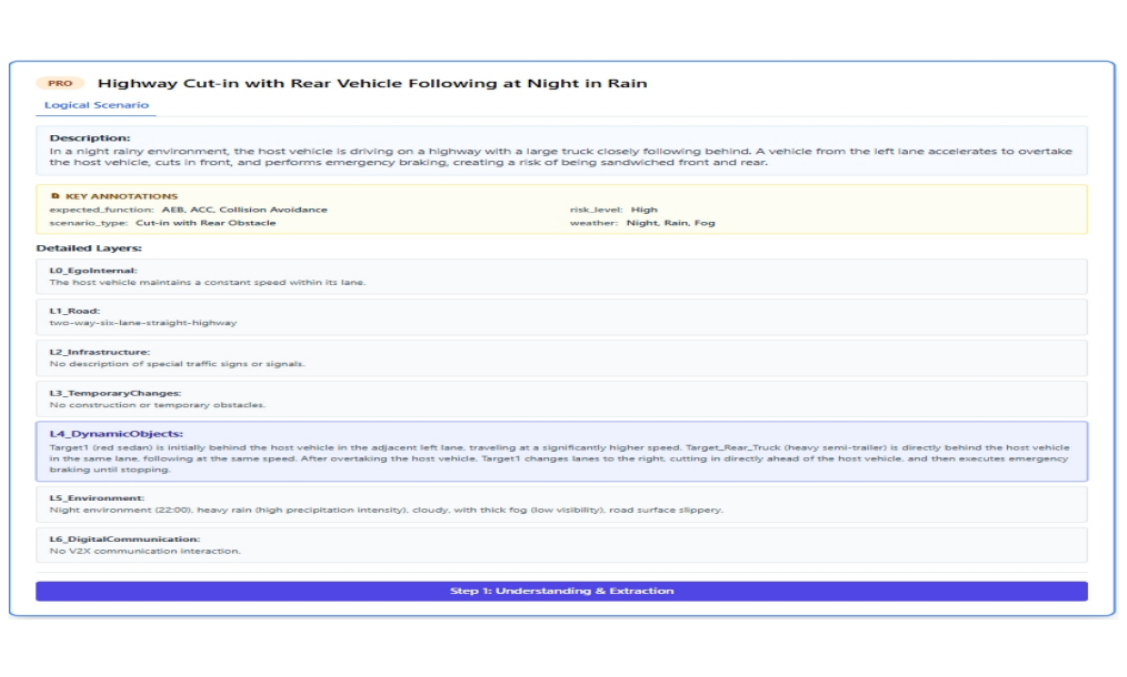
Click Functional Scenario Gen to enter the AI scenario generation function process。


AI Data Engine
Generate Dangerous Scenarios from Test Function
Generate Scenario Description (JSON)

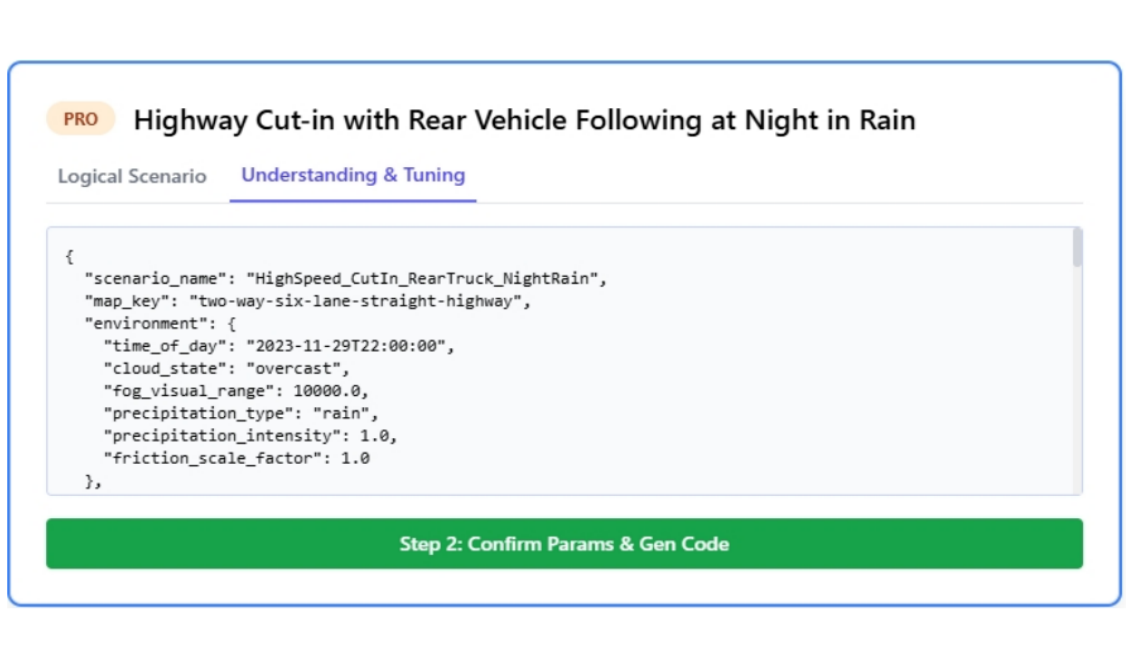
Users can fill in the test function or scenario description based on actual needs. L0 to L6 are also optional. Click Generate Scenario Description (JSON) to generate the logical scenario。


AI Data Engine
Highway Cut-in with Rear Vehicle Following at Night in Rain
Highway Cut-in and Emergency Braking at Night in Rain
Highway Cut-in and Emergency Braking at Night in Rain
Highway Cut-in with Following Large Truck
Highway Cut-in and Braking under Visual Occlusion
Highway Left Lane Overtaking Cut-in with Emergency Braking and Right Lane Parallel Blocking
High-speed aggressive lane change colliding with stationary motorcycle
Highway Right Lane Change Encountering Rear Interfering Vehicle
Left Vehicle Overtaking and Cutting in Followed by Emergency Braking
Aggressive Double Lane Change Cut-in
Highway Left Lane Cut-in with Emergency Braking
Highway Left Lane Cut-in with Emergency Braking

The section ahead displays scenario library scenes recommended by the AI based on the user's description, while the section that follows shows scenes generated by the AI. Click on any scenario card：

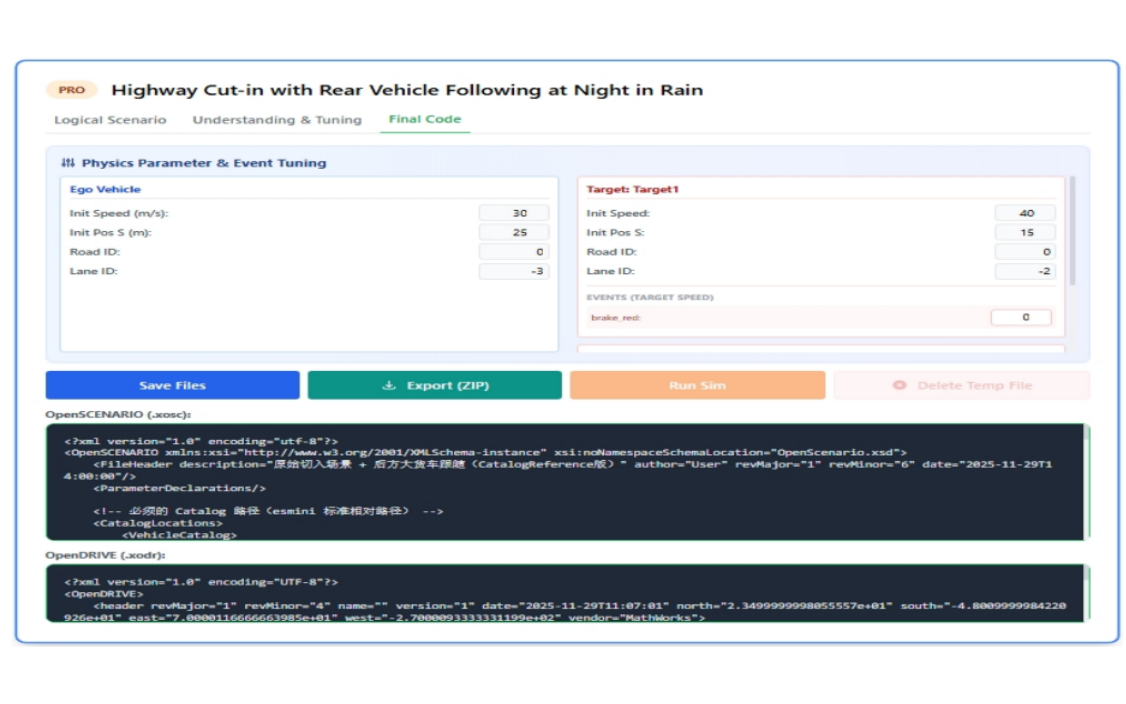

Highway Cut-in with Rear Vehicle Following at Night in Rain
Logical Scenario
Step 1: Understanding & Extraction

Then click Step 1: Understanding & Extraction to generate the corresponding physical scenario:

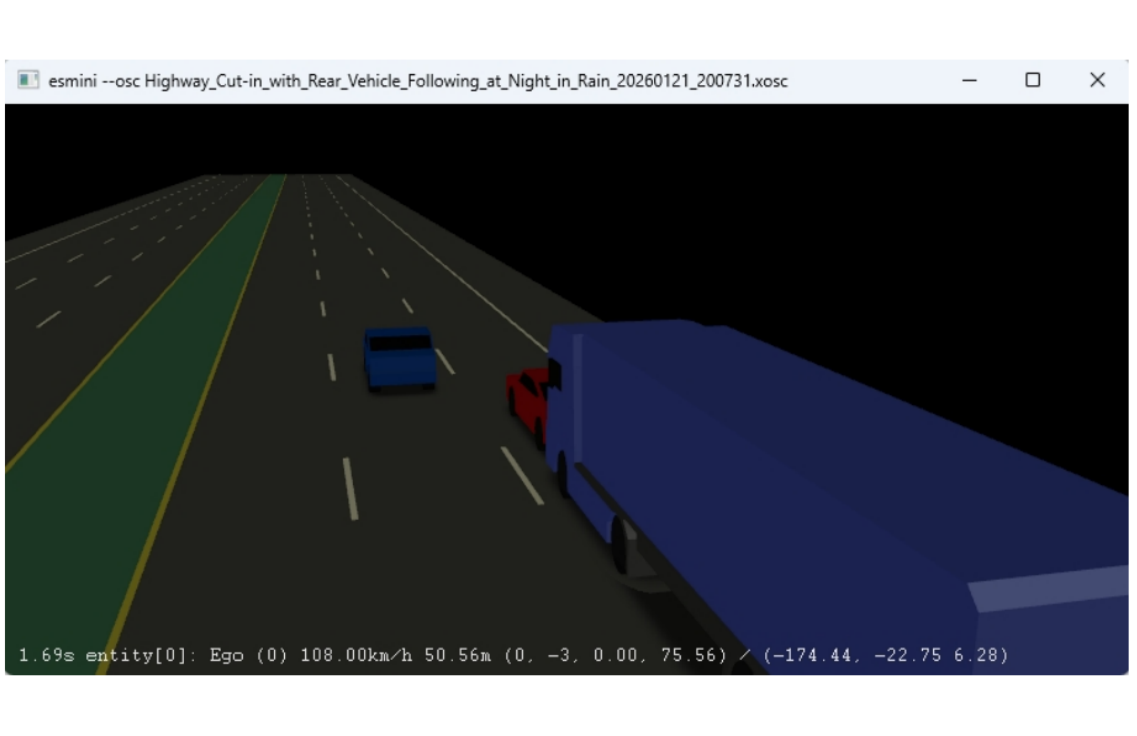

PRO
Highway Cut-in with Rear Vehicle Following at Night in Rain
Logical Scenario
Understanding & Tuning
Step 2: Confirm Params & Gen Code

```
{
  "scenario_name": "HighSpeed_CutIn_RearTruck_NightRain",
  "map_key": "two-way-six-lane-straight-highway",
  "environment": {
    "time_of_day": "2023-11-29T22:00:00",
    "cloud_state": "overcast",
    "fog_visual_range": 10000.0,
    "precipitation_type": "rain",
    "precipitation_intensity": 1.0,
    "friction_scale_factor": 1.0
  },
```

Click Step 2: Confirm Params & Gen Code to generate OS code, which also includes some adjustable parameters that users can modify as needed：

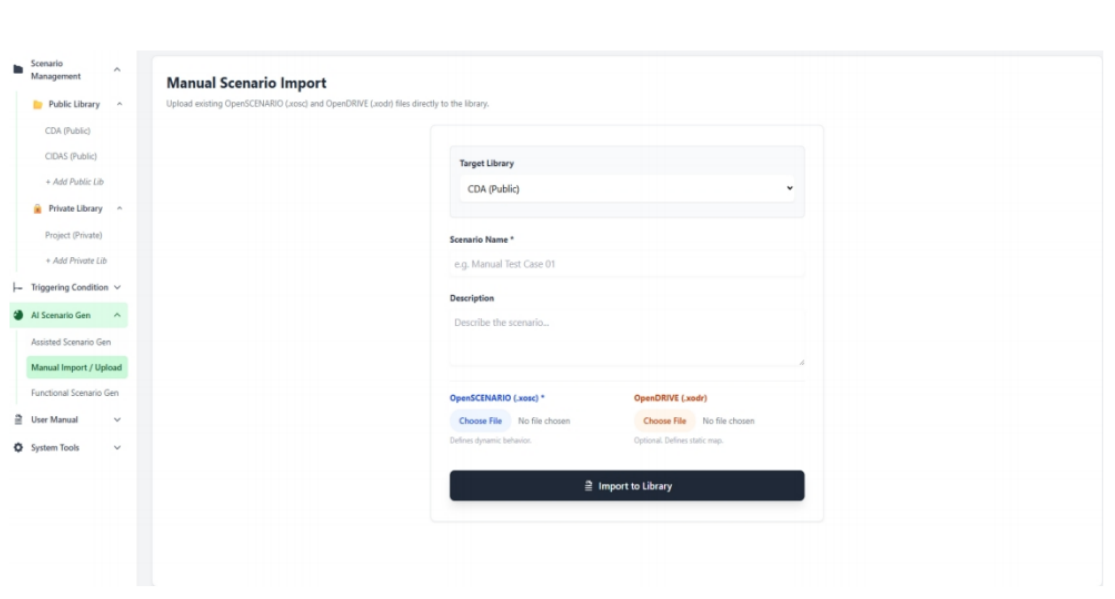

PRO
Highway Cut-in with Rear Vehicle Following at Night in Rain
Logical Scenario
Understanding & Tuning
Final Code
Physics Parameter & Event Tuning
Ego Vehicle
Target: Target1
Save Files
Export (ZIP)

Finally, click Save Files (Apply Params) to save the code, and then click Esmini Simulation to start the simulation：

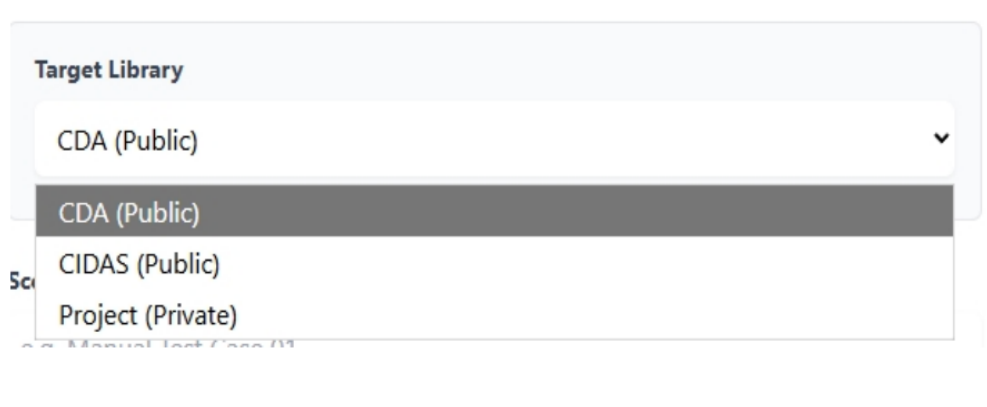

esmini --osc Highway_Cut-in_with_Rear_Vehicle_Following_at_Night_in_Rain_20260121_200731.xosc

（3）Custom Scenario Addition Function Process：

Click Manual Import / Upload to enter the user-defined scenario addition interface。


Manual Scenario Import
Target Library
CDA (Public)
Scenario Name *
Description
Import to Library

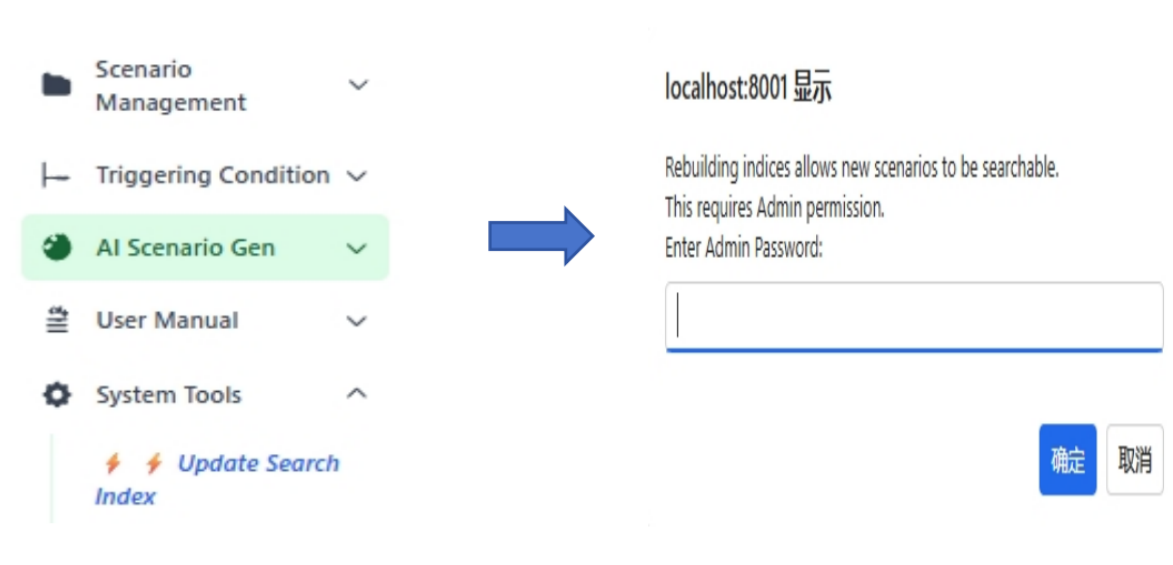
Select the appropriate target storage library as needed：


Target Library
CDA (Public)
CDA (Public)
CIDAS (Public)
Project (Private)

After entering the scenario name and description, upload the corresponding OpenScenario/OpenDrive code file, then click Import to Library to save the scenario。

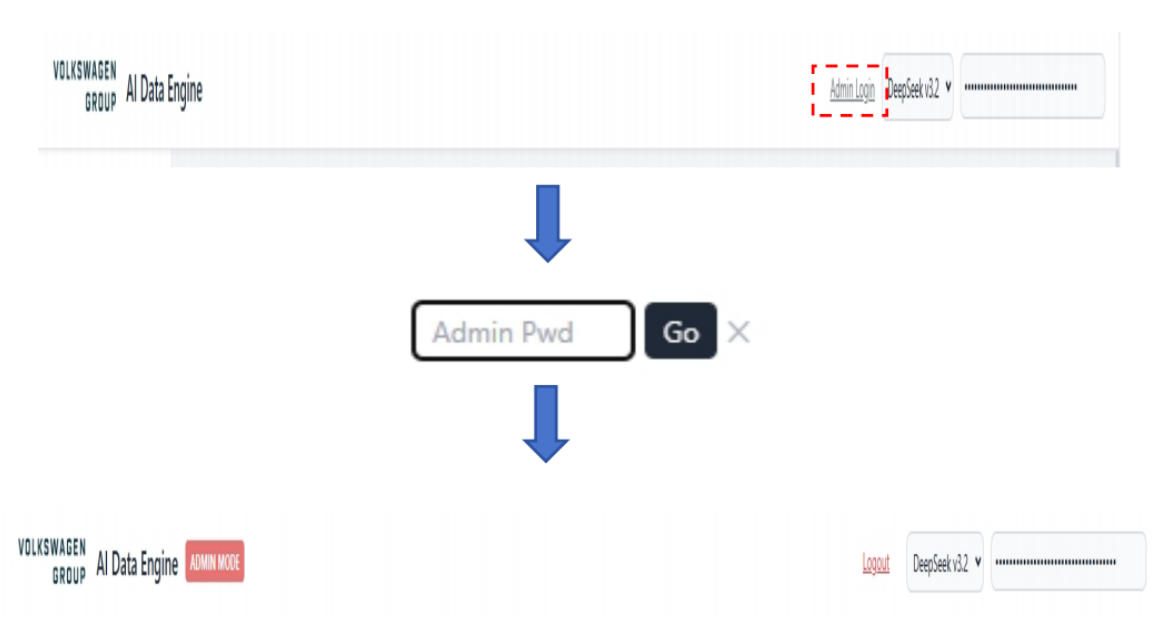
3.System Tools Description

The system tools include an Update Vector Database function. After uploading existing or new scenarios, you need to click Update Search Index to enter the password (admin888), and then proceed to update the vector database. This allows you to search for the desired scenarios in Scenario Management.


Scenario Management
Triggering Condition
AI Scenario Gen
User Manual
System Tools
Update Search Index
localhost:8001 显示
Rebuilding indices allows new scenarios to be searchable.
This requires Admin permission.
Enter Admin Password:
确定
取消

4.Administrator Mode

On the UI interface, click Admin Login and enter the password（admin888) to enter Administrator Mode。


VOLKSWAGEN GROUP
AI Data Engine
Admin Login
DeepSeek v3.2
Admin Pwd
Go
VOLKSWAGEN GROUP
AI Data Engine
ADMIN MODE
Logout
DeepSeek v3.2